

Sequence modelling for multi-event detection

Evaluation against non-neural baselines, and the model deployed

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Summary

The detection rules built in step 4 classify single events. This work evaluated whether a neural sequence model can classify a run of alerts instead. Across 8 held out networks a GRU over the alert sequence scored 0.199 mean average precision against 0.249 for logistic regression on the same windows, and won no fold, so a neural model is not justified at this data scale. A portable logistic model scoring 0.177 is deployed on the dashboard with its measurement recorded alongside it.

1. Dataset

AIT Alert Data Set, Zenodo record 8263181, CC-BY-4.0, published with the CSET 2024 paper. 2,600,263 Wazuh alerts from 8 simulated enterprise networks, each containing a labelled multi-step intrusion, in native `alerts.json` format. Kaggle and HuggingFace held no comparable set of real Wazuh alerts with ground truth phases.

Scope limit. Rules 100100 to 100113 have no parent signature in this dataset, and rules 5501 and 5502 are the only overlap. The evaluation therefore measures the method and not the lab ruleset. Measuring the ruleset requires a campaign against the lab endpoints.

2. Sampling and evaluation protocol

PARAMETER	VALUE
Episode	300 second tumbling window, labelled attack if it overlaps a ground truth phase
Episodes	8,915, of which 187 are attack. Base rate 2.1%
Split	Leave one network out, 8 folds, three seeds per fold for the sequence model. Whole networks are held out rather than the tail of a timeline, and the feature vocabulary is built from training sources only, so a held out network contributes no features
Metric	Average precision. f1 at a fixed cutoff is unstable at this base rate: the same predictions score 0.049 at 0.5 and 0.367 at a validation-selected threshold. Thresholds are selected on validation, never on the test set

3. Models and results

Four models were evaluated under the protocol above. The logistic regression, average precision and metrics are implemented in numpy, approximately sixty lines in total. The sequence model is PyTorch: an embedding over rule identifiers, a GRU, and a linear head.

MODEL	F1	AVERAGE PRECISION
Best single rule, the baseline	0.245 (sd 0.134)	0.161 (sd 0.092)
Logistic regression on shape and severity, deployed	0.192 (sd 0.107)	0.177 (sd 0.080)
GRU over the alert sequence	0.180 (sd 0.112)	0.199 (sd 0.081)
Logistic regression on rule counts and timing	0.292 (sd 0.121)	0.249 (sd 0.097)
Random, for reference	—	0.021, the base rate

Logistic regression on rule counts and timing wins all 8 folds, the GRU none, with a mean margin of 0.049 average precision. All four score above the 0.021 base rate and 0.249 is approximately 12 times base rate at ranking, so the alert stream carries signal. Sequence order is not where it is.

4. Deployed model

The winning model uses one feature column per dataset rule identifier. Applied to this lab every such count is zero, so it cannot be deployed. A portable variant was fitted on 11 vocabulary-independent features describing burst shape and alert severity, which transfer because Wazuh rule levels are a property of the ruleset rather than of a network. It scores 0.177 average precision against 0.249 for the non-portable model, a cost of 0.072 and a retention of 71%. The remaining signal is therefore concentrated in which rules fired rather than in the shape of the burst, which supports the rule-based approach already implemented.

The scorer runs inside the status script the dashboard already sends to the manager on each poll. No additional service, port, dependency or manager-side installation was required. Alerts are bucketed into 300 second windows and each window is scored, at a measured cost of 4 ms against a 90 ms poll budget.

5. Severity scoring

Model output ranks windows by how unusual they are, which is not the same as impact. Each window therefore also receives a 0 to 100 severity score: a weighted composite of the model margin, peak rule level, volume of high-level alerting, burst density, distinct signatures and distinct attack stages. A multiplier of up to 30% applies when credential access and persistence appear in the same window, indicating a chain rather than two independent events.

TEST CASE	SEVERITY
Quiet window, session opens only	16
Brute force burst, rule 100111	54
Brute force burst with local account creation	78

The single-event rules do not distinguish the second case from the third: both events are reported with no relationship recorded between them. Windows scoring 50 or above are recorded as findings, expandable to the per-term working and exportable to PDF. All constants are stated judgements rather than fitted parameters, held in one file and printed beside the score.

6. Limitations and next step

The model is fitted on eight public networks and has not been measured on this lab, whose rules appear in no public dataset. The score is therefore triage ordering rather than a detection, and that statement is recorded in the model file and shown on the dashboard panel. Removing the limitation requires a scenario campaign against the lab endpoints, producing labelled alerts that carry the lab ruleset.

Architecture diagrams are in [docs/architecture/](#). The modelling report and the per-fold measurements are available on request.